Old Memory: Ongoing Asymmetric Risks

April 19, 2026 11:06 PM GMT

We continue to prefer MLC NAND and NOR Flash over DDR4. Within legacy DRAM, we think spec migration and new opportunities differentiate stocks.

Key Takeaways

- MLC NAND/NOR pricing power should accelerate from 2Q26 led by Macronix.
- The DDR4 pricing hike is likely to be up 60% in 2Q, then decelerate into 2H.
- Intel EMIB could drive notable upside on silicon capacitors, mainly from AP Memory.
- Spec migration should continue to be the highlight from PSMC in 2027.
- We are OW Macronix, AP Memory (raised PT), PSMC and GigaDevice (raised PT). We are EW Nanya and Winbond.

Keeping our selective view; need to consider outlook beyond 2026: Old memory stocks typically show higher volatility when the cycle starts to roll forward (see memory downgrade note). We stick to our view that legacy NAND (especially MLC NAND) is better than DDR4. In this note, we also highlight new opportunities for legacy DRAM vendors to tap in the advance packaging area, growing along with rising non GPU server demand (see Shawn Kim's agentic AI report).

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DDR4 - definitely still in tightness , but momentum decelerating: We believe DDR4 8Gb pricing will likely be above USD20 soon (and 4Gb above USD10). Some customers still see strong momentum given pull-ins (e.g., PC). This suggests DDR4 vendors could see high profitability in the near term. That said, we see limited share price upside amid pricing hike deceleration, KV cache compression, server system optimization and LTA discussions. CXMT is also adding some DDR4 support (see GigaDevice note).

Legacy flash - accelerating the pricing hike: We continue to see extreme tightness for MLC NAND, especially with reports of Kioxia phasing out of the market after Samsung (TrendForce News, 31 Mar 2026). We believe MXIC is trying to reallocate capacity to support MLC NAND, resulting in supply reduction on NOR. Thus, we believe both MLC NAND and NOR pricing could be up 100% at most in 2Q, and be maintained at high double digits into 2H.

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New opportunities from EMIB-T: We believe AP Memory will be the major EMIB-T proxy in Taiwan, as it is the provider of silicon capacitors. Amid high power consumption from ASIC (e.g., TPU), we think its content per chip could be near USD100. This could be significant considering potential future customers (see our EMIB report). Incremental revenue could be 50%+ vs. the 2026e level. We think the most important factor to focus on is prepayment from some end-customers.

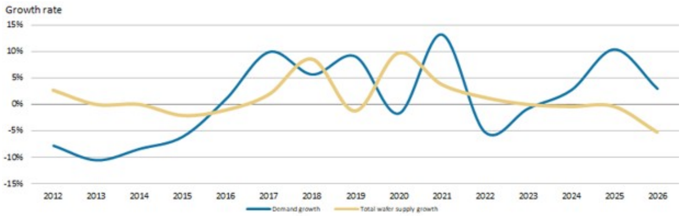
Key Charts and Exhibits

Exhibit 1: Order of preference: GC memory

	Macronix 2337.TW	AP Memory 6531.TW	GigaDevice 603986.SS	PSMC 6770.TW	Winbond 2344.TW
Rating	Overweight	Overweight	Overweight	Overweight	Equal-Weight
Trading Currency	TWD	TWD	CNY	TWD	TWD
Price Target	**202.0**	**777.0**	**348.0**	**71.0**	**100.0**
Current Price	131.0	644.0	289.0	52.6	87.8
Upside/(Downside) (%)	54%	21%	20%	35%	14%
Market Cap (in USD mm)	7,513.9	3,312.1	28,293.1	6,920.0	11,663.9
Avg Daily Traded Vol (in USD mm)	135.5	48.1	871.0	158.1	281.7
Street View: Ratings					
Buy/Overweight	90%	100%	92%	40%	78%
Hold/Equal-weight	0%	0%	8%	60%	22%
Sell/Underweight	10%	0%	0%	0%	0%
Bull Case Value	326.0	1,145.0	604.0	113.6	120.0
Upside (%)	149%	78%	109%	116%	37%
Bear Case Value	119.5	376.0	219.0	35.5	41.4
Downside (%)	-9%	-42%	-24%	-33%	-53%
Risk/Reward Skew	17.0	1.9	4.5	3.6	0.7
Morgan Stanley Estimates					
FY26e	TWD	TWD	CNY	TWD	TWD
Sales	56,677	9,333	17,061	66,303	224,249
EBITDA	20,949	2,901	6,005	12,145	106,288
EBIT	13,412	2,841	5,736	4,581	90,677
EPS	5.90	16.16	8.10	8.61	16.57
FY27e					
Sales	109,328	13,777	24,555	72,058	264,897
EBITDA	48,356	4,905	9,504	18,971	129,416
EBIT	43,644	4,842	9,235	11,931	109,239
EPS	19.61	26.32	12.98	2.21	19.87
FY26 MSe vs. Consensus Mean					
Sales	-18.0%	15.2%	18.0%	4.7%	8.3%
EBITDA	-6.7%	13.9%	2.3%	-26.1%	5.8%
EBIT	-41.6%	25.6%	11.1%	-33.2%	6.8%
EPS	-44.9%	-2.7%	11.2%	28.0%	7.7%
FY27 MSe vs. Consensus Mean					
Sales	2.0%	13.2%	35.4%	4.8%	-1.0%
EBITDA	-0.6%	17.6%	35.6%	-9.4%	-1.8%
EBIT	27.4%	19.5%	41.7%	5.6%	3.2%
EPS	-23.7%	14.5%	40.0%	-9.9%	-3.4%
Valuation Multiples at Last Close					
FY26e					
P/E	22.2x	39.9x	35.7x	6.1x	5.3x
EV/EBIT	20.7x	32.6x	31.3x	18.1x	4.3x
EV/EBITDA	13.2x	31.9x	29.9x	6.8x	3.7x
EV/Sales	4.9x	9.9x	10.5x	1.2x	1.7x
FCF Yield	-12.0%	1.8%	2.3%	54.9%	10.2%
FY27e					
P/E	6.7x	24.5x	22.3x	23.8x	4.4x
EV/EBIT	5.6x	18.8x	18.8x	6.8x	2.6x
EV/EBITDA	5.0x	18.5x	18.3x	4.3x	2.2x
EV/Sales	2.2x	6.6x	7.1x	1.1x	1.1x
FCF Yield	15.5%	3.2%	4.0%	6.0%	32.5%

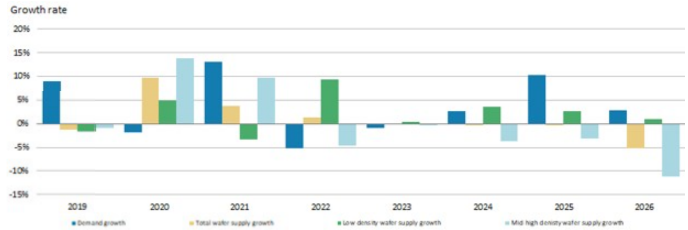
Source: FactSet, Morgan Stanley Research (e) estimates. Pricing as of April 17, 2026.

Exhibit 2: NOR flash demand and supply growth rates



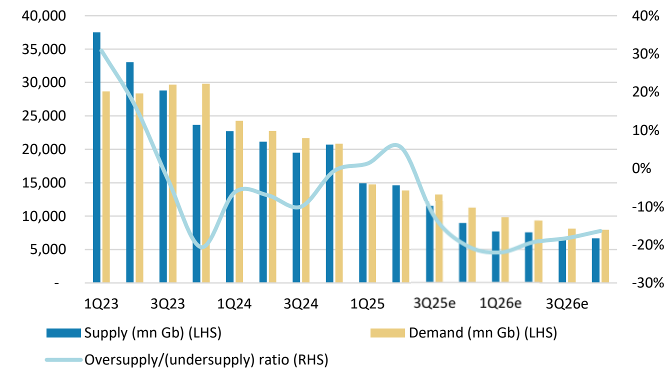
Source: Company data, Morgan Stanley Research

Exhibit 3: NOR flash demand growth and supply growth by density



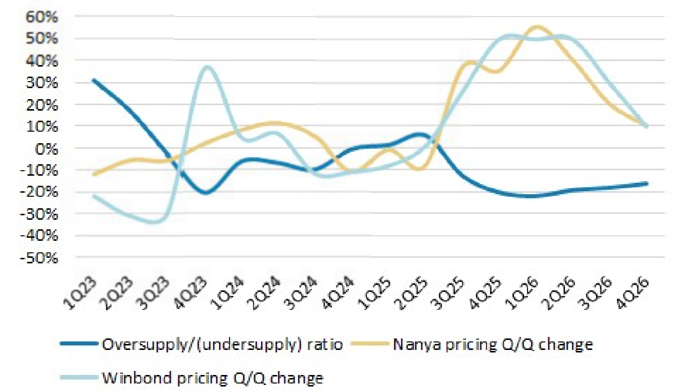
Source: Company data, Morgan Stanley Research

Exhibit 4: DDR4 quarterly supply and demand summary



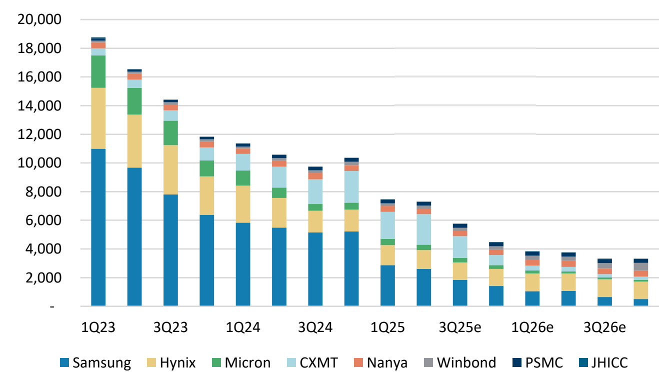
Source: Morgan Stanley Research (e) estimates

Exhibit 5: Quarterly oversupply/undersupply ratio vs. Nanya and Winbond pricing Q/Q change



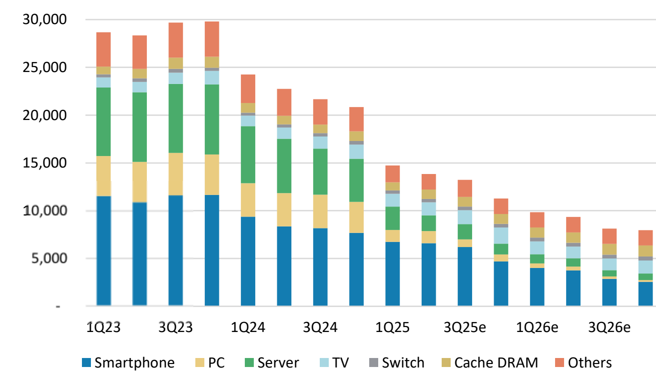
Source: Morgan Stanley Research (e) estimates

Exhibit 6: Quarterly supply breakdown (mn Gb)



Source: Morgan Stanley Research (e) estimates

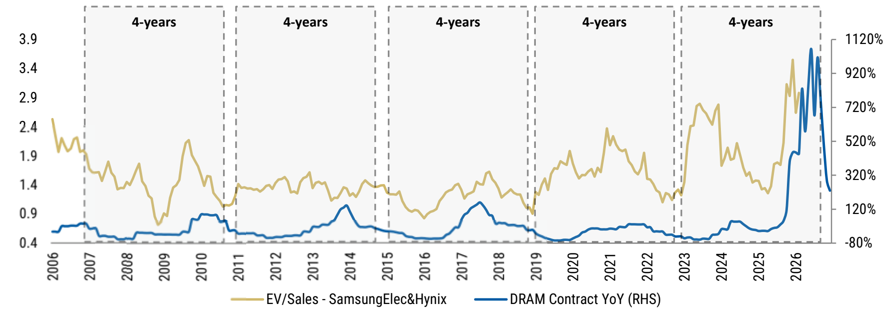
Exhibit 7: Quarterly demand breakdown by product (mn Gb)



Source: Morgan Stanley Research (e) estimates

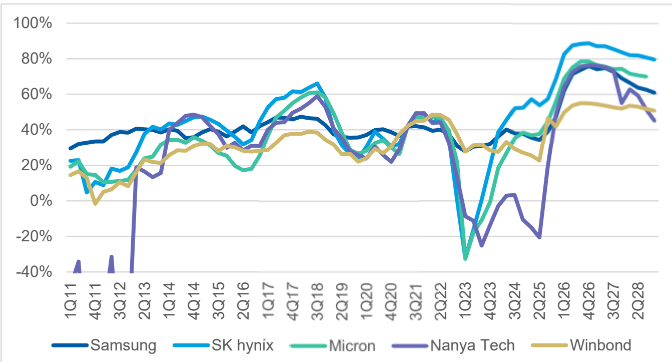
Exhibit 8:

EV/sales vs. DRAM price YoY



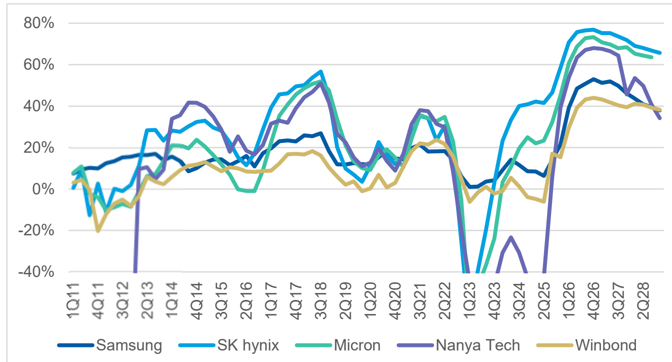
Source: FactSet, Morgan Stanley Research. Future DRAM contract Y/Y based on Morgan Stanley Research estimates.

Exhibit 9: Memory vendors GM



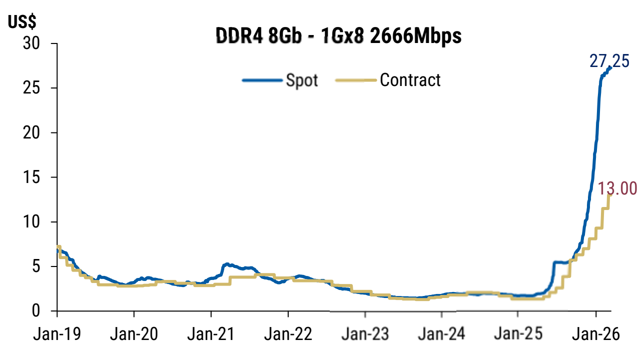
Source: Company data, Morgan Stanley Research estimates

Exhibit 10: Memory vendors OPM



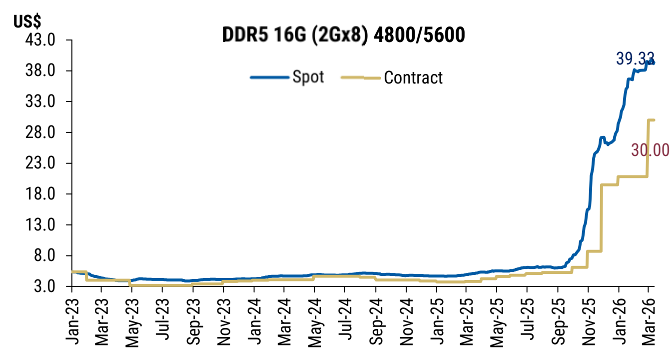
Source: Company data, Morgan Stanley Research

Exhibit 11: DDR4 8Gb (1Gx8) pricing chart



Source: DRAMeXchange, Morgan Stanley Research; data as of Mar 13, 2026

Exhibit 12: DDR5 16Gb (2Gx8) pricing chart



Source: DRAMeXchange, Morgan Stanley Research; data as of Mar 13, 2026

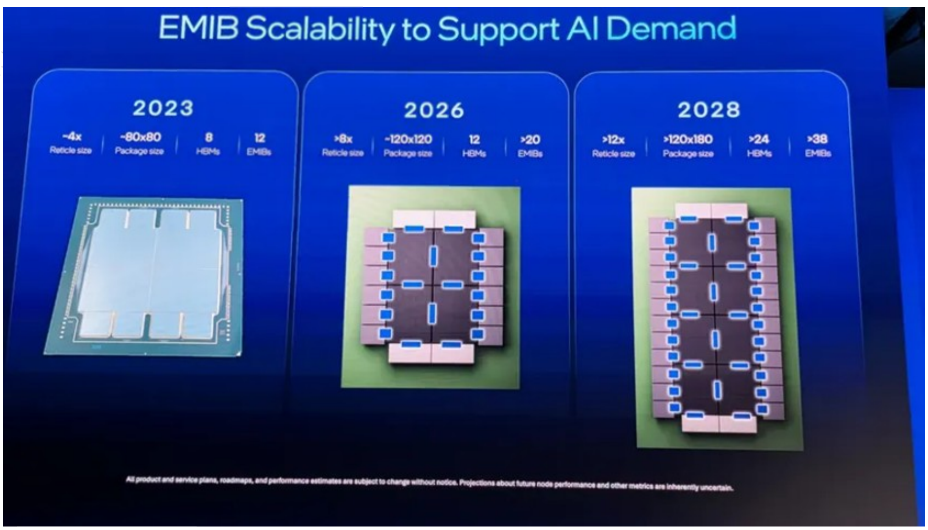
Intel EMIB Opportunity for AP Memory

Intel's (covered by Morgan Stanley Research Analyst Joe Moore) Embedded Multi-die Interconnect Bridge (EMIB) takes a more modular and cost-efficient approach: Instead of a full interposer, it uses small embedded silicon bridges only where high-bandwidth connections are needed, while relying on the organic substrate for the rest. Theoretically, this improves yield and reduces cost, while allowing flexible chiplet-based designs across different process nodes.

The downside is that EMIB does not provide a uniform high-density interconnect across the entire package, which can limit total bandwidth and make system-level routing more complex. As a result, it is generally less optimal for extreme bandwidth use cases compared to CoWoS, but it can support larger reticle-size design.

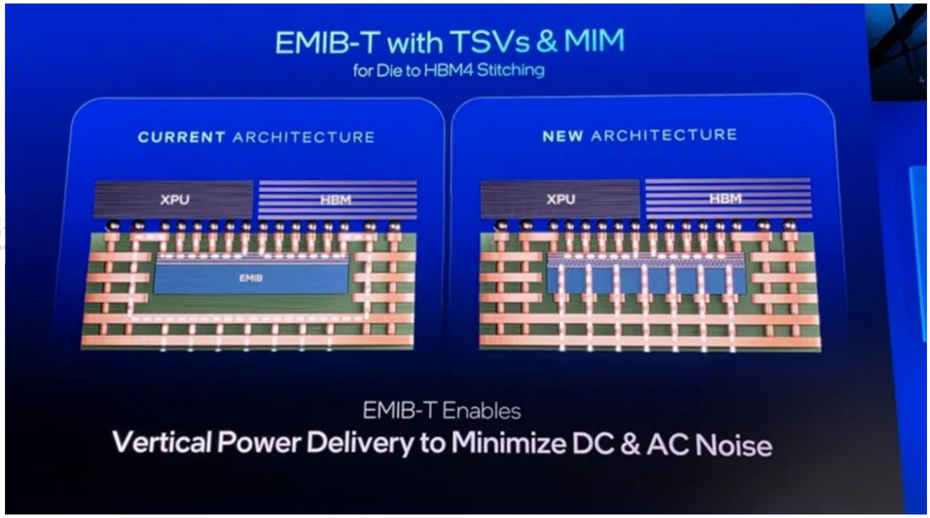
Also, besides EMIB technology itself, our latest industry checks suggest that those chip designers are concerned about the yield or execution from Intel since it lacks of the experience of mass produce EMIB-T products as well as serving external clients. Therefore, compared to TSMC, the operational risk and ultimate yield would also be taken into consideration from those customers to allocate how much chip production to Intel EMIB-T.

Exhibit 13: Intel's EMIB can support larger chips with more reticles (>12) if its supply chain can execute well



Source: Intel

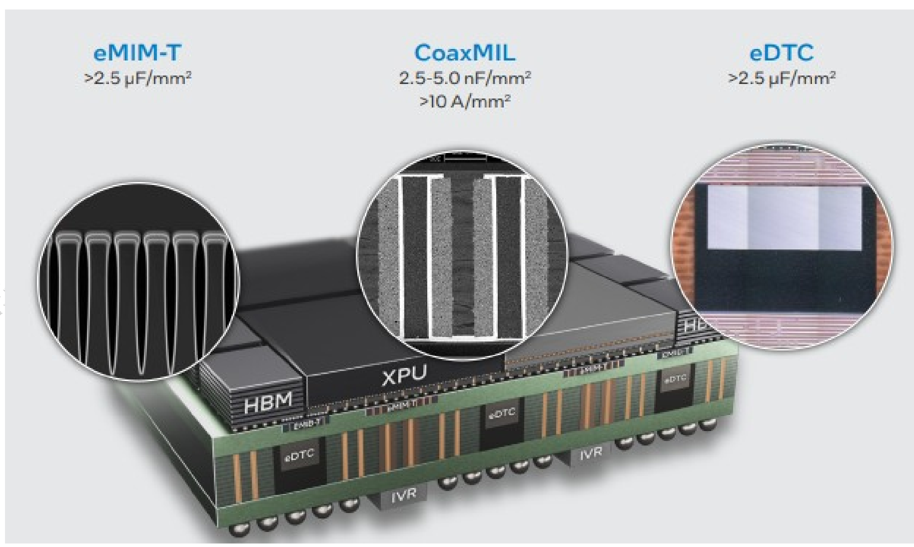
Exhibit 14: Intel's EMIB-T (new architecture) embeds silicon bridges featuring vertical copper connections (TSVs) into organic substrates, enabling superior power delivery, reduced noise, and high-speed HBM4 memory integration



Source: Intel

According to Intel, to further stabilize power delivery, Intel Foundry integrates embedded deep trench capacitors (eDTC) for ultra-high capacitance density close to the load, thereby flattening PDN impedance across the base die. Embedded MIM (eMIM-T) technology delivers superior decoupling at the base die, leveraging the same advanced architecture as Omni MIM even on mature nodes. For precise, high-speed point-of-load regulation, CoaxMIL technology embeds magnetic inductors into the substrate, enabling integrated voltage regulators (IVRs) with rapid response and minimal loss. Together, these technologies form a scalable power delivery fabric that keeps pace with rising compute density. We believe these eDTC are IPD products that AP Memory provides.

Exhibit 15: An example of Intel Foundry's power delivery solution shown at Intel Direct Connect 2025, showcasing forward-looking plans for increased eMIM-T capacity; we think this is adopting AP Memory's IPD, which is the eDTC here



Source: Intel

For Intel EMIB's customer front, our checks suggest that both Google and AWS have reached out to Intel's EMIB-T for mechanical testing vehicles (MTV); we believe the production could be related to MediaTek's next-generation 2nm version TPU as well as Alchip's next generation 2nm Trainium4, while the trial run is likely to be in 2027 and mass production to be in 2028 if everything goes smoothly. Since Alchip already has a strong relationship with Intel on Gaudi production, we believe Alchip has been more familiar collaborating with Intel and its workflow.

We still do not yet know how those next-generation chips will be allocated between Intel's EMIB-T vs. TSMC's CoWoS/CoPoS. However, both Google and AWS are the two leaders among major CSPs in terms of AI ASIC design. We therefore see these two companies needing larger chip size design in their ASIC roadmap; the allocation will depend greatly on 1) Intel's EMIB-T production yield, 2) TSMC's CoWoS reticle support, and potentially 3) TSMC's CoPoS schedule.

Besides Google and AWS, our checks also suggest that Meta is approaching Intel for potential EMIB-T production for its future AI ASIC as well. However, the progress may be at an even earlier stage compared to Google and AWS, in our view, because its ASIC is less mature than other CSPs.

For further details, see: Greater China Semiconductors: The Foundry Floorplan: TSMC preview update; CoWoS vs. EMIB competition (14 Apr 2026).

AP Memory: Estimate Revision Summary

We raise our 2026/2027/2028 EPS estimates by 2%/13%/13%, respectively: We factor in 1Q26 preliminary sales, which we see leading to higher 2026 earnings. We raise increase our 2026-27 revenue forecasts due to the stronger IPD outlook and a better pricing environment for the IoT business from memory price hike. In comparison, consensus 2026 and 2027 EPS estimates are NT$14.52 and NT$22.98, respectively. Our stronger EPS forecasts reflect our more constructive view on the IPD shipment outlook, we believe.

Exhibit 16: AP Memory – Estimate revisions

(NT$ mn)	New '26e	Old '26e	Diff.	New '27e	Old '27e	Diff.	New '28e	Old '28e	Diff.
Net sales	9,333	8,572	9%	13,777	11,640	18%	18,865	16,678	13%
COGS	4,784	4,175		6,724	5,449		8,896	7,833	
Gross profit	4,550	4,397	3%	7,053	6,191	14%	9,969	8,845	13%
Operating expenses	1,709	1,625		2,211	1,927		2,913	2,659	
Operating profit	2,841	2,772	2%	4,842	4,265	14%	7,056	6,187	14%
Non-op. income (exp.)	343	343		343	343		343	343	
Pretax income	3,183	3,115	2%	5,185	4,607	13%	7,399	6,529	13%
Taxes	556	544		906	805		1,293	1,141	
Net income	2,627	2,570	2%	4,279	3,802	13%	6,106	5,388	13%
Reported EPS	**16.16**	**15.81**	**2%**	**26.32**	**23.39**	**13%**	**37.56**	**33.14**	**13%**
Margins									
Gross margin	48.7%	51.3%	-2.6%	51.2%	53.2%	-2.0%	52.8%	53.0%	-0.2%
Operating margin	30.4%	32.3%	-1.9%	35.1%	36.6%	-1.5%	37.4%	37.1%	0.3%
Pretax margin	34.1%	36.3%	-2.2%	37.6%	39.6%	-1.9%	39.2%	39.1%	0.1%
Net margin	28.1%	30.0%	-1.8%	31.1%	32.7%	-1.6%	32.4%	32.3%	0.1%

Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 17: AP Memory – Quarterly financial summary

YE DEC31 (NT$ mn)	1Q26e	2Q26e	3Q26e	4Q26e	1Q27e	2Q27e	3Q27e	4Q27e	1Q28e	2Q28e	3Q28e	4Q28e	2025	2026e	2027e	2028e
Total Revenues	**2,101**	**2,222**	**2,451**	**2,558**	**2,938**	**3,168**	**3,650**	**4,021**	**4,276**	**4,566**	**4,916**	**5,108**	**5,658**	**9,333**	**13,777**	**18,865**
Sequential Change	*12.5%*	*5.8%*	*10.3%*	*4.4%*	*14.8%*	*7.8%*	*15.2%*	*10.2%*	*6.3%*	*6.8%*	*7.7%*	*3.9%*				
Change vs Year Ago	*115.5%*	*67.3%*	*64.0%*	*37.0%*	*39.9%*	*42.5%*	*48.9%*	*57.2%*	*45.5%*	*44.1%*	*34.7%*	*27.0%*	*35.0%*	*65.0%*	*47.6%*	*36.9%*
Cost of Sales	1,098	1,145	1,248	1,293	1,456	1,562	1,765	1,941	2,025	2,158	2,316	2,397	3,024	4,784	6,724	8,896
Percent of Revenues	*52%*	*52%*	*51%*	*51%*	*50%*	*49%*	*48%*	*48%*	*47%*	*47%*	*47%*	*47%*	*53%*	*51%*	*49%*	*47%*
Gross Profit	**1,003**	**1,078**	**1,204**	**1,265**	**1,482**	**1,606**	**1,885**	**2,081**	**2,250**	**2,408**	**2,600**	**2,711**	**2,634**	**4,550**	**7,053**	**9,969**
Percent of Revenues	*47.7%*	*48.5%*	*49.1%*	*49.5%*	*50.4%*	*50.7%*	*51.6%*	*51.7%*	*52.6%*	*52.7%*	*52.9%*	*53.1%*	*46.5%*	*48.7%*	*51.2%*	*52.8%*
Incremental Margin	*31%*	*61%*	*55%*	*57%*	*57%*	*54%*	*58%*	*53%*	*67%*	*54%*	*55%*	*58%*	*33%*	*52%*	*56%*	*57%*
Total Opex	401	411	439	458	499	504	577	631	667	708	757	781	1,234	1,709	2,211	2,913
Percent of Revenues	*19.1%*	*18.5%*	*17.9%*	*17.9%*	*17.0%*	*15.9%*	*15.8%*	*15.7%*	*15.6%*	*15.5%*	*15.4%*	*15.3%*	*21.8%*	*18.3%*	*16.0%*	*15.4%*
Operating Income	**602**	**667**	**765**	**807**	**982**	**1,103**	**1,308**	**1,449**	**1,583**	**1,700**	**1,843**	**1,929**	**1,399**	**2,841**	**4,842**	**7,056**
Percent of Revenues	*28.6%*	*30.0%*	*31.2%*	*31.6%*	*33.4%*	*34.8%*	*35.8%*	*36.0%*	*37.0%*	*37.2%*	*37.5%*	*37.8%*	*24.7%*	*30.4%*	*35.1%*	*37.4%*
Total Non-operating Income (Loss)	86	86	86	86	86	86	86	86	86	86	86	86	119	343	343	343
Profit Before Taxes	687	752	851	893	1,068	1,188	1,394	1,535	1,669	1,786	1,929	2,015	1,518	3,183	5,185	7,399
Percent of Revenues	*33%*	*34%*	*35%*	*35%*	*36%*	*38%*	*38%*	*38%*	*39%*	*39%*	*39%*	*39%*	*27%*	*34%*	*38%*	*39%*
Taxes	120	131	149	156	187	208	244	268	292	312	337	352	279	556	906	1,293
Tax Rate	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*17.5%*	*18.3%*	*17.5%*	*17.5%*	*17.5%*
Reported Income (TW GAAP)	**567**	**621**	**702**	**737**	**881**	**981**	**1,150**	**1,267**	**1,377**	**1,474**	**1,592**	**1,663**	**1,240**	**2,627**	**4,279**	**6,106**
Percent of Revenues	*27.0%*	*27.9%*	*28.6%*	*28.8%*	*30.0%*	*31.0%*	*31.5%*	*31.5%*	*32.2%*	*32.3%*	*32.4%*	*32.6%*	*21.9%*	*28.1%*	*31.1%*	*32.4%*
Change vs Year Ago	*71%*	*-214%*	*-1%*	*-4%*	*55%*	*58%*	*64%*	*72%*	*56%*	*50%*	*38%*	*31%*	*-21%*	*112%*	*63%*	*43%*
Reported EPS (NT$, TW GAAP)	**3.49**	**3.82**	**4.32**	**4.53**	**5.42**	**6.03**	**7.07**	**7.79**	**8.47**	**9.07**	**9.79**	**10.23**	**7.63**	**16.16**	**26.32**	**37.56**
Change vs Year Ago	*71%*	*-214%*	*-1%*	*-4%*	*55%*	*58%*	*64%*	*72%*	*56%*	*50%*	*38%*	*31%*	*-22%*	*112%*	*63%*	*43%*

Source: Company data, Morgan Stanley Research (e) estimates

AP Memory: Valuation Methodology

We raise our price target to NT$777 from NT$607 to factor in preliminary sales for 1Q26 and our estimate changes for 2026-28. We also lift our intermediate growth rate from 11% to 12.8% as we factor in the Intel EMIB opportunity for AI in the long run.

Aside from intermediate growth, our other residual income model assumptions are unchanged, including: 1) cost of equity unchanged at 9.2% (2.0% risk-free rate, 6% risk premium, 1.2 beta); 2) payout ratio of 67%; and 3) terminal growth rate of 3% to be more in line with the long-term macro growth rate. Our new PT implies 2026e P/E of 48x and 2027e P/E of 30x – more than 1 s.d. above the historical average since 2020. Our bull and bear case values rise to NT$1,145 (from NT$895) and NT$376 (from NT$294), respectively.

Exhibit 18: AP Memory: Residual Income model

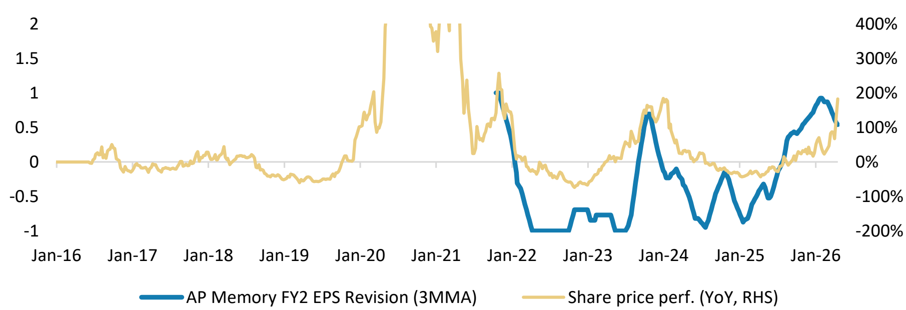
(NT$ mn)	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e
Total Equity	**14,156**	**16,666**	**19,890**	**22,623**	**25,706**	**29,182**	**33,102**	**37,523**	**42,508**	**48,130**	**54,470**	**61,620**
Net Profit	**2,627**	**4,279**	**6,106**	**6,885**	**7,765**	**8,756**	**9,874**	**11,135**	**12,557**	**14,161**	**15,969**	**18,009**
ROAE	**19.8%**	**27.8%**	**33.4%**	**32.4%**	**32.1%**	**31.9%**	**31.7%**	**31.5%**	**31.4%**	**31.2%**	**31.1%**	**31.0%**
Residual Income	1,311	2,627	4,033	4,612	5,187	5,836	6,567	7,392	8,321	9,370	10,553	11,886
Spread	**10.6%**	**18.6%**	**24.2%**	**23.2%**	**22.9%**	**22.7%**	**22.5%**	**22.3%**	**22.2%**	**22.0%**	**21.9%**	**21.8%**
Ending Equity Capital	14,156											
PV of Forecast Period	37,328											
PV of Continuing Value	74,916											
Equity Value	**126,400**											
No. of Shares	163											
Projected Price (NT$)	**777**											

Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 19: AP Memory: Historical forward P/E

Source: Company data, Morgan Stanley Research

Exhibit 20: AP Memory: Earnings estimate revision breadth



Source: Company data, Morgan Stanley Research estimates

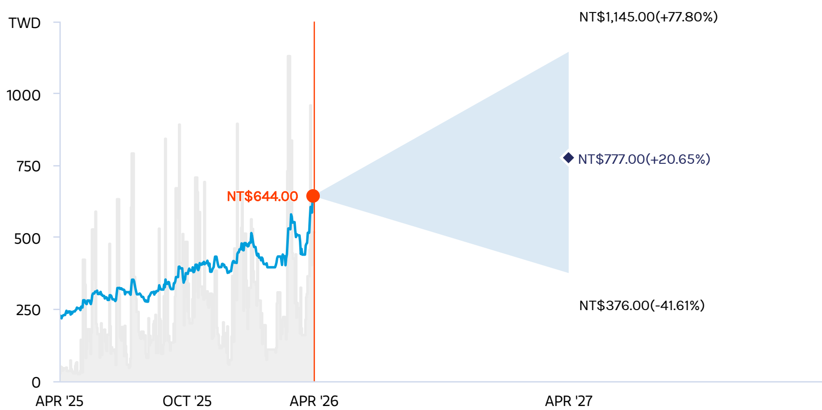
Risk Reward – AP Memory Technology Corp (6531.TW)

OW on IPD and WoW opportunities

PRICE TARGET NT$777.00

Base case, derived from a residual income model. Key RI model assumptions include 1) cost of equity constant at 9.2% (2.0% risk-free rate, 6% risk premium, 1.2 beta); 2) payout ratio of 67%; 3) terminal growth rate of 3.0%; and 4) medium-term growth rate of 12.8%.

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- IPD (interposers and discretes) revenue is likely to ramp in 2026-27, and be further driven by new IPD interposer opportunities into 2028.
- Memory bandwidth and power consumption are the next issues to be resolved in AI computing, and we believe AP Memory's wafer-on wafer (WoW) packaging technology could be an effective solution.
- The company expects memory business to be stable in the near term.
- Overall AI business should continue to rise in the mix – a positive read for the margin profile.
- Our PT implies 48x 2026e P/E and 30x 2027e P/E - roughly 1 s.d. above the average level since 2020.

Consensus Rating Distribution

● 100% Overweight
0% Equal-weight
0% Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*
Technology Diffusion: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$1,145.00

71x 2026e EPS

Faster-than-expected AI and IPD business progress with PSRAM generating more cash flow: We assume: 1) WoW packaging revenue to reach >NT$10bn, and IPD revenue to reach <NT$8bn by 2028; 2) total revenue to rise at a 55%+ CAGR, 2025-28; and 3) gross margin to increase from 46.5% in 2025 to 60%+ by 2028.

BASE CASE NT$777.00

48x 2026e EPS

Fast-growing AI and IPD business, with PSRAM generating stable cash flows: We expect: 1) WoW revenue to reach NT$5.5bn, and IPD revenue to reach NT$7.6bn by 2028; 2) total revenue to rise at a 49% CAGR, 2025-28; and 3) gross margin to increase from 46.5% in 2025 to over 52% in 2028.

BEAR CASE NT$376.00

23x 2026e EPS

Slower-than-expected AI and IPD business progress, with PSRAM facing intensifying competition: We assume: 1) WoW packaging revenue of ~NT$1.5bn, and IPD revenue of ~NT$2bn by 2028; 2) total revenue to rise at a 10-15% CAGR, 2025-28; and 3) gross margin to remain at ~51% by 2028.

Risk Reward – AP Memory Technology Corp (6531.TW)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
1. Wifi Sales (%) (%)	73.7	55.7	40.8	30.7
2. Ethernet Sales (%) (%)	7.1	8.7	21.3	29.0

INVESTMENT DRIVERS

- Demand for AI applications
- Consumer tech demand, especially from the IoT market

GLOBAL REVENUE EXPOSURE

- 0-10% India
- 0-10% Japan
- 0-10% Latin America
- 0-10% MEA
- 0-10% North America
- 10-20% APAC, ex Japan, Mainland China and India
- 10-20% Europe ex UK
- 30-40% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger-than-expected consumer demand
- Faster-than-expected IPD ramp and WoW packaging development
- Milder competition from partners (foundry, memory house, GPU vendor) or other design houses

RISKS TO DOWNSIDE

- Weaker-than-expected consumer demand
- Slower-than-expected IPD ramp and WoW packaging development
- Intensifying competition from partners (foundry, memory house, GPU vendor) or other design houses

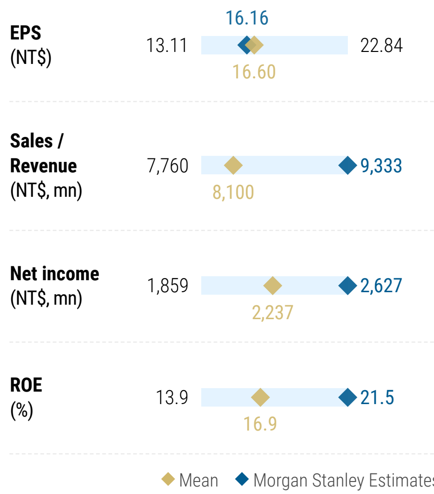
OWNERSHIP POSITIONING

Inst. Owners, % Active 56.2%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

AP Memory: Financial Summary

Exhibit 21: Financial summary

Income Statement

NT$mn (Years End Dec)	2025	2026e	2027e	2028e
Net sales	5,658	9,333	13,777	18,865
COGS	(3,024)	(4,784)	(6,724)	(8,896)
Gross profit	2,634	4,550	7,053	9,969
Operating expenses	(1,234)	(1,709)	(2,211)	(2,913)
Operating income	**1,399**	**2,841**	**4,842**	**7,056**
Non-operating income	119	343	343	343
Pre-tax income	1,518	3,183	5,185	7,399
Income tax	56	153	249	355
Net Income bf. emp. bonus	1,575	3,336	5,434	7,754
Employee bonus expense	335	709	1,155	1,649
Reported net Income	**1,240**	**2,627**	**4,279**	**6,106**
Adj.wtd.avg.shrs (million)	163	163	163	163
Reported EPS (NT$)	**7.63**	**16.16**	**26.32**	**37.56**
Modelware Diluted EPS (NT$)	**7.63**	**16.16**	**26.32**	**37.56**

Balance Sheet

NT$mn (Years End Dec)	2025	2026e	2027e	2028e
Cash	7,263	8,263	9,877	12,086
Mkt Securities	3,756	3,756	3,756	3,756
AR/NR	588	1,026	1,514	2,074
Inventory	1,145	1,546	2,173	2,875
Other	187	187	187	187
Current Assets	**12,939**	**14,778**	**17,507**	**20,977**
Long-term investments	1,206	1,206	1,206	1,206
Fixed assets	57	63	69	76
Deffered assets	126	126	126	126
Other assets	411	411	411	411
Total Assets	**14,740**	**16,584**	**19,320**	**22,797**
S/T borrowings	200	200	200	200
AP/NP	505	557	783	1,037
Other ST liabilities	1,582	1,582	1,582	1,582
LT debt	0	0	0	0
Other LT liabilities	89	89	89	89
Common shares	814	814	814	814
Total Liabilities	**2,376**	**2,428**	**2,654**	**2,907**
Additional capital	7,595	7,595	7,595	7,595
Retained earning	3,776	5,568	8,078	11,302
Other shareholders' equity	179	179	179	179
Total Equity	**12,364**	**14,156**	**16,666**	**19,890**
Total Liab. & Shrhldr's Equity	**14,740**	**16,584**	**19,320**	**22,797**

e = Morgan Stanley Research Estimates

Source: Morgan Stanley Research, Company Data

Cash Flow Statement

NT$mn (Years End Dec)	2025	2026e	2027e	2028e
Cashflow from Operations	**1,575**	**1,901**	**3,452**	**5,162**
Net profits	1,240	2,627	4,279	6,106
Depreciation	59	61	62	64
Working Capital Change	(145)	(787)	(889)	(1,008)
Other adjustments	421	0	0	0
Cashflow from Investing	**(67)**	**(66)**	**(69)**	**(71)**
Capex	(26)	(66)	(69)	(71)
Change of LT Investment	0	0	0	0
Change of ST Investment	0	0	0	0
Other adjustments	(41)	0	0	0
Cashflow from financing	**1,564**	**(835)**	**(1,769)**	**(2,881)**
Increase in L/T debt	0	0	0	0
Increase in S/T debt	100	0	0	0
Cash Dividend Paid	(968)	(835)	(1,769)	(2,881)
Dir& Emp Bonus Paid	0	0	0	0
Issuance of stock	0	0	0	0
Other adjustments	2,433	0	0	0
Exchange rate adjustment	3	0	0	0
Net change in cash	**3,075**	**1,000**	**1,614**	**2,209**

Financial Ratios

	2025	2026e	2027e	2028e
Growth(%)				
Turnover	35.0	65.0	47.6	36.9
Operating profits	31.6	103.0	70.5	45.7
Pretax profits	-25.3	109.6	62.9	42.7
Net profits	-21.4	111.9	62.9	42.7
EPS	-21.6	111.8	62.9	42.7
Margins (%)				
Gross Margin	46.5	48.7	51.2	52.8
Operating Margin	24.7	30.4	35.1	37.4
Pretax Margin	26.8	34.1	37.6	39.2
Net Profit	21.9	28.1	31.1	32.4
Return (%)				
ROAE	10.2	19.8	27.8	33.4
ROAA	8.9	16.8	23.8	29.0
Gearing (%)				
Net Debt/Equity	(57.1)	(57.0)	(58.1)	(59.8)
Liabilities/Equity	19.2	17.2	15.9	14.6
Ratios (X)				
Current ratio	5.7	6.3	6.8	7.4
Quick ratio	3.4	4.0	4.4	5.0
Others				
AR/NR Turnover (days)	40	40	40	40
Inventory Turnover (days)	118	118	118	118
AP Turnover (days)	43	43	43	43
Cash Conversion (days)	116	116	116	116

GigaDevice: Estimate Revisions and Quarterly Financials

We increase our 2025 EPS estimate by 2%, essentially maintain 2026, and lift 2027 by 13%: We reflect preliminary 4Q25 revenue, leading to a slight increase on the 2025 bottom line. We lift our 2027 revenue forecast as we believe GigaDevice will ramp up 16G LPDDR4 and 32G DDR4 at CXMT's Gen 4 platform in 2027, based on its latest announcement. GM should rise slightly in 2027 amid a better product mix.

Exhibit 22: GigaDevice - Estimate revisions

Rmb mn	2025E	2025E	Diff.	2026E	2026E	Diff.	2027E	2027E	Diff.
Net sales	9,203	9,203	0%	17,061	17,061	0%	24,555	22,608	9%
COGS	5,502	5,619		8,509	8,509		12,045	11,253	
Gross profit	3,701	3,584	3%	8,552	8,552	0%	12,510	11,355	10%
Operating expenses	2,176	2,079		2,816	2,816		3,275	3,177	
Operating profit	1,526	1,505	1%	5,736	5,736	0%	9,235	8,177	13%
Non-op. income (exp.)	181	171		100	100		100	100	
Pretax Income	1,706	1,676	2%	5,836	5,836	0%	9,335	8,277	13%
Taxes	29	36		175	175		280	248	
Net income	1,648	1,610	2%	5,628	5,626	0%	9,022	7,994	13%
Reported EPS (Rmb)	**2.48**	**2.42**	**2%**	**8.10**	**8.09**	**0%**	**12.98**	**11.50**	**13%**
Margins									
Gross margin	40.2%	38.9%	1.3%	50.1%	50.1%	0.0%	50.9%	50.2%	0.7%
Operating margin	16.6%	16.4%	0.2%	33.6%	33.6%	0.0%	37.6%	36.2%	1.4%
Pretax margin	18.5%	18.2%	0.3%	34.2%	34.2%	0.0%	38.0%	36.6%	1.4%
Net margin	17.9%	17.5%	0.4%	33.0%	33.0%	0.0%	36.7%	35.4%	1.4%

Source: Company data, Morgan Stanley Research (E) estimates

Exhibit 23: GigaDevice: Quarterly financials

Rmb mn	1Q25	2Q25	3Q25	4Q25p	1Q26E	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025E	2026E	2027E
Total Revenues	**1,909**	**2,241**	**2,681**	**2,372**	**3,246**	**4,012**	**4,663**	**5,140**	**5,289**	**5,893**	**6,521**	**6,852**	**7,356**	**9,203**	**17,061**	**24,555**
Sequential Change	*11.9%*	*17.4%*	*19.6%*	*-11.5%*	*36.8%*	*23.6%*	*16.2%*	*10.2%*	*2.9%*	*11.4%*	*10.7%*	*5.1%*				
Change vs Year Ago	*17.3%*	*13.1%*	*31.4%*	*39.0%*	*70.0%*	*79.0%*	*73.9%*	*116.7%*	*62.9%*	*46.9%*	*39.9%*	*33.3%*	*27.7%*	*25.1%*	*85.4%*	*43.9%*
Cost of Sales	1,194	1,412	1,589	1,307	1,775	2,029	2,274	2,431	2,535	2,880	3,237	3,394	4,561	5,502	8,509	12,045
Percent of Revenues	*63%*	*63%*	*59%*	*55%*	*55%*	*51%*	*49%*	*47%*	*48%*	*49%*	*50%*	*50%*	*62%*	*60%*	*50%*	*49%*
Gross Profit	**715**	**830**	**1,092**	**1,065**	**1,471**	**1,983**	**2,389**	**2,709**	**2,753**	**3,013**	**3,285**	**3,458**	**2,795**	**3,701**	**8,552**	**12,510**
Percent of Revenues	*37.4%*	*37.0%*	*40.7%*	*44.9%*	*45.3%*	*49.4%*	*51.2%*	*52.7%*	*52.1%*	*51.1%*	*50.4%*	*50.5%*	*38.0%*	*40.2%*	*50.1%*	*50.9%*
Incremental Margin	*73%*	*35%*	*60%*	*NM*	*46%*	*67%*	*62%*	*67%*	*30%*	*43%*	*43%*	*53%*	*51%*	*49%*	*62%*	*53%*
Total Opex	540	545	574	518	624	678	730	784	747	793	844	891	1,984	2,176	2,816	3,275
Percent of Revenues	*28.3%*	*24.3%*	*21.4%*	*21.8%*	*19.2%*	*16.9%*	*15.7%*	*15.3%*	*14.1%*	*13.5%*	*12.9%*	*13.0%*	*27.0%*	*23.6%*	*16.5%*	*13.3%*
R&D	292	276	292	257	290	300	315	335	290	300	315	335	1,122	1,117	1,240	1,240
Percent of Revenues	*15.3%*	*12.3%*	*10.9%*	*10.8%*	*8.9%*	*7.5%*	*6.8%*	*6.5%*	*5.5%*	*5.1%*	*4.8%*	*4.9%*	*15.3%*	*12.1%*	*7.3%*	*5.0%*
General & administrative	136	156	161	160	172	177	182	192	193	198	203	213	491	612	723	807
Percent of Revenues	*7.1%*	*6.9%*	*6.0%*	*6.7%*	*5.3%*	*4.4%*	*3.9%*	*3.7%*	*3.6%*	*3.4%*	*3.1%*	*3.1%*	*6.7%*	*6.7%*	*4.2%*	*3.3%*
Selling & marketing	112	113	121	101	162	201	233	257	264	295	326	343	371	446	853	1,228
Percent of Revenues	*5.8%*	*5.0%*	*4.5%*	*3.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*5.0%*	*4.8%*	*5.0%*	*5.0%*
Operating Income	**175**	**285**	**518**	**547**	**847**	**1,305**	**1,659**	**1,925**	**2,006**	**2,221**	**2,441**	**2,568**	**811**	**1,526**	**5,736**	**9,235**
Percent of Revenues	*9.2%*	*12.7%*	*19.3%*	*23.1%*	*26.1%*	*32.5%*	*35.6%*	*37.4%*	*37.9%*	*37.7%*	*37.4%*	*37.5%*	*11.0%*	*16.6%*	*33.6%*	*37.6%*
Change vs Year Ago	*12.7%*	*19.4%*	*36.9%*	*1331.9%*	*383.2%*	*357.9%*	*220.3%*	*251.7%*	*136.9%*	*70.1%*	*47.1%*	*33.4%*	*130%*	*88%*	*276%*	*61%*
Total Non-operating Income(Loss)	60	75	10	35	25	25	25	25	25	25	25	25	313	181	100	100
Profit Before Taxes	**236**	**360**	**528**	**583**	**872**	**1,330**	**1,684**	**1,950**	**2,031**	**2,246**	**2,466**	**2,593**	**1,124**	**1,706**	**5,836**	**9,335**
Percent of Revenues	*12%*	*16%*	*20%*	*25%*	*27%*	*33%*	*36%*	*38%*	*38%*	*38%*	*38%*	*38%*	*15%*	*19%*	*34%*	*38%*
Taxes	(4)	12	11	10	26	40	51	58	61	67	74	78	23	29	175	280
Tax Rate	*-1.7%*	*3.4%*	*2.1%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*3.0%*	*2.0%*	*1.7%*	*3.0%*	*3.0%*
Net Income, Cont Ops	240	348	517	573	846	1,290	1,633	1,891	1,970	2,178	2,392	2,515	1,101	1,677	5,661	9,055
Percent of Revenues	*13%*	*16%*	*19%*	*24%*	*26%*	*32%*	*35%*	*37%*	*37%*	*37%*	*37%*	*37%*	*15%*	*18%*	*33%*	*37%*
Reported Income	**235**	**341**	**508**	**565**	**838**	**1,282**	**1,625**	**1,883**	**1,962**	**2,170**	**2,383**	**2,507**	**1,103**	**1,648**	**5,628**	**9,022**
Percent of Revenues	*12%*	*15%*	*19%*	*24%*	*26%*	*32%*	*35%*	*37%*	*37%*	*37%*	*37%*	*37%*	*15%*	*18%*	*33%*	*37%*
Change vs Year Ago	*15%*	*9%*	*61%*	*109%*	*257%*	*276%*	*220%*	*233%*	*134%*	*69%*	*47%*	*33%*	*584%*	*49%*	*242%*	*60%*
Reported EPS (Rmb)	**0.35**	**0.51**	**0.76**	**0.85**	**1.20**	**1.84**	**2.34**	**2.71**	**2.82**	**3.12**	**3.43**	**3.61**	**1.66**	**2.48**	**8.10**	**12.98**
Change vs Year Ago	*15%*	*10%*	*61%*	*108%*	*241%*	*259%*	*207%*	*220%*	*134%*	*69%*	*47%*	*33%*	*585%*	*50%*	*227%*	*60%*

Source: Company data, Morgan Stanley Research (E) estimates

GigaDevice: Valuation methodology

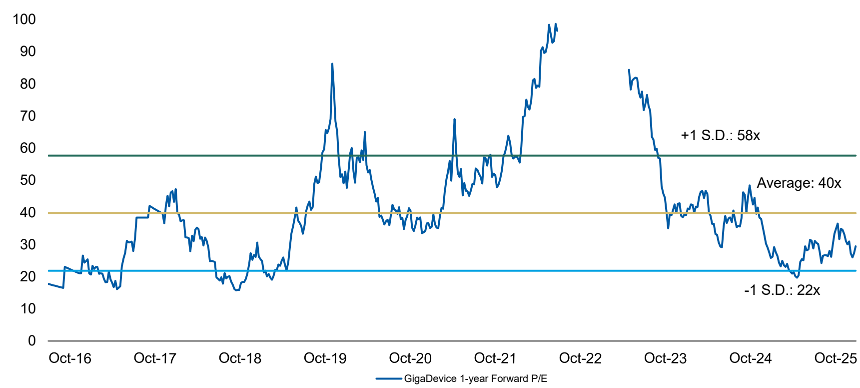
We increase our price target from Rmb301 to Rmb348: This reflect our higher earnings estimates and an increase in our intermediate growth rate assumption from 16.0% to 16.4%, mainly on the new opportunity in LPDDR4 in 2027. We continue to derive our price target from a residual income model, which we believe best captures the stock's long-term value. Our key assumptions include: cost of equity of 8.9% (beta 1.1, risk-free rate 2%, and risk premium 6%), payout ratio of 40% and a terminal growth rate of 3.0%. We reduce our medium-term and terminal growth rates estimates to reflect our more cautious view on DDR4. Our new price target implies 2026e P/E of 43x, roughly in line with the historical average of 40x since 2016. We increase our bull case value from Rmb523 to Rmb604 and our bear case value from Rmb219 to Rmb254.

Exhibit 24: GigaDevice: Residual Income model

Rmb mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total Equity	**23,234**	**30,568**	**37,268**	**45,063**	**54,132**	**64,684**	**76,962**	**91,247**	**107,867**	**127,205**	**149,705**
Net Profit	**5,628**	**9,022**	**10,498**	**12,214**	**14,211**	**16,534**	**19,238**	**22,383**	**26,043**	**30,301**	**35,255**
ROAE	**27.2%**	**33.5%**	**31.0%**	**29.7%**	**28.7%**	**27.8%**	**27.2%**	**26.6%**	**26.2%**	**25.8%**	**25.5%**
Residual Income	3,321	5,728	6,744	7,746	8,907	10,255	11,822	13,643	15,760	18,222	21,086
Spread	**18.3%**	**24.7%**	**22.1%**	**20.8%**	**19.8%**	**18.9%**	**18.3%**	**17.7%**	**17.3%**	**16.9%**	**16.6%**
Ending Equity Capital	23,234										
PV of Forecast Period	61,039										
PV of Continuing Value	157,466										
Equity Value	**241,740**										
No. of Shares	695										
Projected Price (Rmb)	**348**										

Source: Morgan Stanley Research (E) estimates

Exhibit 25: GigaDevice: historical forward P/E



Source: Company data, Morgan Stanley Research estimates

Risk Reward – GigaDevice Semiconductor Beijing Inc (603986.SS)

Multiple drivers ahead

PRICE TARGET Rmb348.00

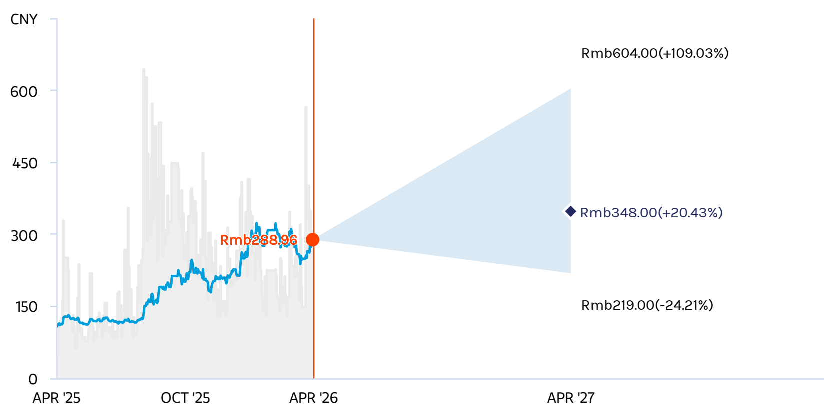
Our price target of Rmb301 is our base case value, derived from our residual income model. Key assumptions: cost of equity of 8.9% (beta 1.1, risk-free rate 2% and risk premium 6%), a payout ratio of 40%, medium-term growth rate of 16.4%, and a terminal growth rate of 3.0%.

Consensus Price Target Distribution Rmb141.00 — Rmb302.21 — MS PT — Rmb382.00

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect GigaDevice to continue to gain share in China's MCU market. Auto MCUs offer bigger opportunities as local self-sufficiency is low.
- We see further NOR Flash pricing upside, and expect GigaDevice NOR quality to catch up in the auto segment.
- Specialty DRAM pricing started to rebound in late 1Q25, and there are opportunities such as wafer-on-wafer specialty memory.
- We believe GigaDevice will ramp up 16G LPDDR4 and 32G DDR4 at CXMT's Gen 4 platform in 2027.
- Our new price target implies 2026e P/E of 43x, roughly in line with the historical average of 40x since 2016.

Consensus Rating Distribution

● 88% Overweight
12% Equal-weight
0% Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb604.00

75x 2026e EPS

NOR flash prices rise more sharply into 1H26, and the company increases production of mid-/high-density NOR; MCU growth accelerates; DRAM business shows significant contribution: 1) NOR flash sales grow 120%+ Y/Y in 2026 via price hikes and meaningful growth in demand; 2) MCU sales growth significantly exceeds 40% in 2026; 3) contribution from WoW tops expectations in 2026-27e; 4) gross margin above 50% in 2026.

BASE CASE Rmb348.00

43x 2026e EPS

NOR and DRAM pricing upside in 2026; moderate WoW contribution: 1) NOR flash sales rise 60%+ Y/Y in 2026; 2) MCU grows 34% in 2026; 3) moderate contribution from WoW in 2026-27e; 4) gross margin at ~50% in 2026.

BEAR CASE Rmb219.00

31x 2026e EPS

NOR Flash price falls sharply into 1H26; MCU faces stronger competition from foreign and local vendors; pause in DRAM business development: 1) NOR flash sales decline in 2026, as ASP starts to drop; 2) MCU sales decrease >10% Y/Y in 2026; 3) contribution from WoW falls below expectations in 2026-27e; 4) gross margin below 35% in 2026.

Risk Reward – GigaDevice Semiconductor Beijing Inc (603986.SS)

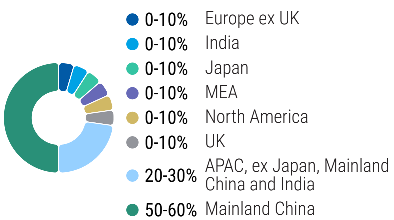
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
NOR Flash ASP growth (%)	(10)	3	46	8
MCU ASP growth (%)	(31)	(8)	0	0
Gross margin (%)	38	40	50	51

INVESTMENT DRIVERS

- Beneficiary of MCU localization
- Decreasing opex burden
- Development of AI glasses market

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

4/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- NOR up-cycle driven by stronger demand
- Superior chip design, leading to increasing exposure to low-density NOR
- Faster-than-expected DRAM development

RISKS TO DOWNSIDE

- NOR down-cycle driven by weaker demand
- Inferior chip design, leading to increasing exposure to mid-/high-density NOR
- Slower-than-expected DRAM development

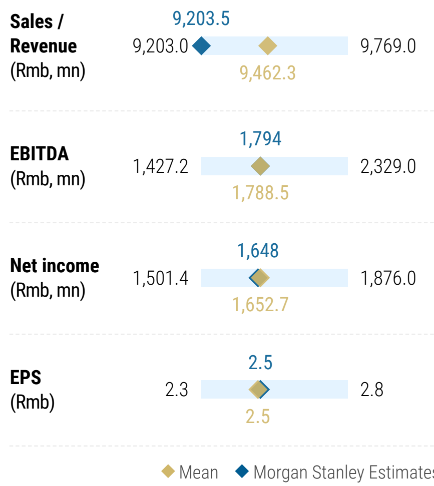
OWNERSHIP POSITIONING

Inst. Owners, % Active 72.3%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2025e



Source: Refinitiv, Morgan Stanley Research

GigaDevice: Financial summary

Income Statement

Rmb mn (Years End Dec)	2024	2025E	2026E	2027E
Net sales	7,356	9,203	17,061	24,555
COGS	(4,561)	(5,502)	(8,509)	(12,045)
Gross profit	2,795	3,701	8,552	12,510
Operating expenses	(1,984)	(2,176)	(2,816)	(3,275)
Operating income	**811**	**1,526**	**5,736**	**9,235**
Non-operating income	313	181	100	100
Pre-tax income	1,124	1,706	5,836	9,335
Income tax	23	29	175	280
Minority interests	2	(29)	(33)	(33)
Reported net Income	**1,103**	**1,648**	**5,628**	**9,022**
Adj.wtd.avg.shrs(m)	666	665	695	695
MW EPS (Rmb)	1.66	2.48	8.10	12.98
Reported EPS (Rmb)	**1.66**	**2.48**	**8.10**	**12.98**

Balance Sheet

Rmb mn (Years End Dec)	2024	2025E	2026E	2027E
Cash	9,128	10,298	14,218	20,167
Mkt Securities	120	120	120	120
AR/NR	232	225	416	599
Inventory	2,346	2,616	4,046	5,728
Other	609	609	609	609
Current Assets	**12,435**	**13,868**	**19,409**	**27,222**
Long-term investments	3,503	3,503	3,503	3,503
Fixed assets	1,062	1,062	1,062	1,062
Deferred assets	426	426	426	426
Other assets	1,803	1,803	1,803	1,803
Total Assets	**19,229**	**20,662**	**26,203**	**34,016**
S/T borrowings	898	898	898	898
AP/NP	734	745	1,152	1,631
Other ST liabilities	699	699	699	699
LT debt	0	0	0	0
Other LT liabilities	220	220	220	220
Total Liabilities	**2,550**	**2,562**	**2,969**	**3,448**
Common shares	664	664	664	664
Additional capital	8,656	8,656	8,656	8,656
Retained earning	6,796	8,218	13,352	20,686
Other shareholders' equity	563	563	563	563
Total Equity	**16,679**	**18,100**	**23,234**	**30,568**
Total Liab. & Shrhldr's Equity	**19,229**	**20,662**	**26,203**	**34,016**

E = Morgan Stanley Research Estimates
Source: Morgan Stanley Research, Company Data

Cash Flow Statement

Rmb mn (Years End Dec)	2024	2025E	2026E	2027E
Cashflow from Operations	**2,032**	**1,666**	**4,683**	**7,906**
Net profits	1,103	1,648	5,628	9,022
Depreciation	269	269	269	269
Working Capital Change	(397)	(251)	(1,214)	(1,386)
Other adjustments	1,058	0	0	0
Cashflow from Investing	**(669)**	**(269)**	**(269)**	**(269)**
Capex	(499)	(269)	(269)	(269)
Change of LT Investment	(1,975)	0	0	0
Change of ST Investment	0	0	0	0
Other adjustments	1,805	0	0	0
Cashflow from financing	**480**	**(226)**	**(494)**	**(1,688)**
Increase in L/T debt	0	0	0	0
Increase in S/T debt	898	0	0	0
Cash Dividend Paid	(15)	(226)	(494)	(1,688)
Dir& Emp Bonus Paid	0	0	0	0
Issuance of stock	5	0	0	0
Other adjustments	(407)	0	0	0
Exchange rate adjustment	107	0	0	0
Net change in cash	**1,950**	**1,170**	**3,920**	**5,948**

Financial Ratios

	2024	2025E	2026E	2027E
Growth(%)				
Turnover	27.7	25.1	85.4	43.9
Operating profits	129.9	88.1	276.0	61.0
EPS	585.3	49.6	226.8	60.3
Margins (%)				
Gross Margin	38.0	40.2	50.1	50.9
Operating Margin	11.0	16.6	33.6	37.6
Pretax Margin	15.3	18.5	34.2	38.0
Net Margin	15.0	17.9	33.0	36.7
Return (%)				
ROAE	6.9	9.5	27.2	33.5
ROAA	6.2	8.3	24.0	30.0
Gearing (%)				
Net Debt/Equity	(49.3)	(51.9)	(57.3)	(63.0)
Liabilities/Equity	15.3	14.2	12.8	11.3
Ratios (X)				
Current ratio	5.3	5.9	7.1	8.4
Quick ratio	4.0	4.5	5.3	6.4
Others				
AR/NR Turnover (days)	9	9	9	9
Inventory Turnover (days)	174	174	174	174
AP Turnover (days)	49	49	49	49
Cash Conversion (days)	133	133	133	133

Valuation Methodology and Risks

Micron Technology Inc. (MU.O)

~25x through-cycle earnings of US $21.00, a premium to history reflecting new opportunities in AI, At the high end of broader semis.

Risks to Upside

- Customers continue to demonstrate an appetite to take on inventory around macroeconomic uncertainty
- Additional wafer intensity of HBM further improves overall supply and demand
- MIcron's HBM share surpasses expectations

Risks to Downside

- Pricing can turn quickly; a falter in end demand with inventories elevated could lead to a swift price reduction
- HBM demand falters and competition intensifies, pressuring pricing

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf